Overview

The general design of this study is based on the online product preference task developed by Powell et al. (2017). On each trial participants will be shown on-screen depictions of pairs of online “products” with respective mean online quality ratings and the sample size on which each mean is based. Participants are asked to make a preference rating for the products. The mean quality rating of the product with the larger number of reviews (hereafter the “more-reviewed product”, MRP) and the discrepancy between the mean quality ratings for the MRP and the less-reviewed product will be manipulated within subjects. The explanation provided for the difference between the sample sizes of the product reviews will be manipulated between groups. The ‘no explanation’ group will receive no explanation for this difference (as per Powell et al., 2017). The ‘explanation’ group will be instructed that the larger number of reviews for the MRP arose because that product had been on the market for a longer time or because it had received celebrity endorsement.

Method

Design and Procedure. The experiment is coded in jsPsych and will be presented to participants via the Amazon Mechanical Turk (MTurk) platform. 25 target items have been constructed by crossing mean quality ratings of the more-reviewed product (2.7, 3.1, 3.8, 4.2, 4.6) with the discrepancy between the value of these ratings and those of the less-reviewed product (+0.3, +0.1, 0, -0.1, and -0.3). For each target item, the two presented products will differ in number of reviews by a constant of 125, but with displayed review numbers varying across the following ranges: more-reviewed product (146-155), less reviewed product (21-30). Figure 1 shows as an example of the way ratings and samples sizes for each rating will be presented (the trial shown is the MRP rating = 3.8, MRP advantage = -0.1).

Additionally, we have constructed four filler items and four attention check items. The fillers present product pairs with different mean ratings but the same number of reviews.

They are included to disguise the systematic variation in mean product values and review sample sizes. Two attention check items will show product pairs where the rating of the MRP is well above that of the LRP (e.g., +1.9 points). The other two check items will be “extreme disadvantage” pairs where the mean rating of the MRP is well below that of the LRP (e.g., -1.9 points). It is assumed that participants who understand task instructions and attend to the task should always choose the MRP for the first pair of check items and the LRP for the second pair. Those who fail both items for either type of check will be excluded from data analyses.

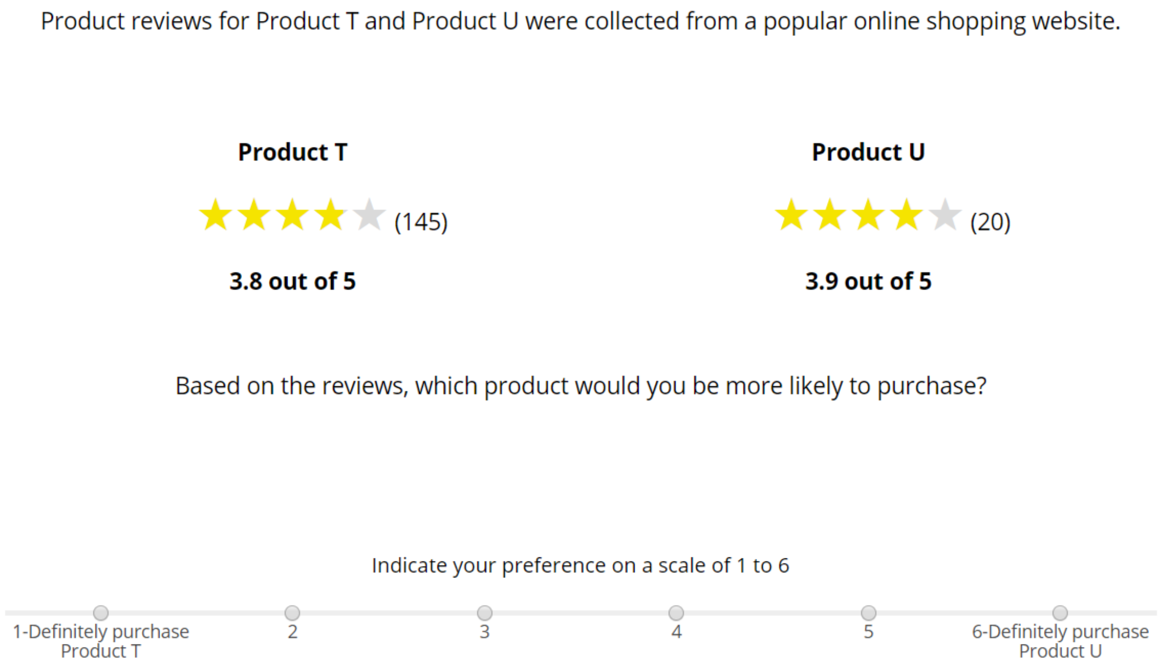

Figure 1. Example of an experimental trial that will be administered via jsPsych.

On each target trial those in the explanation condition will be presented with one of the following two explanations: a) “*The product with the higher number of reviews has been on the market longer than the product with fewer reviews*”; b) “*The product with the higher number of reviews received a celebrity endorsement from a popular lifestyle guru*. These explanations will appear at the bottom of the screen below the target each target item. Which explanation appears on a given trial will be determined randomly using jsPysch code, with the constraint that the two kinds of explanation have to appear on (approximately) equal numbers of target trials. No such explanation will be provided in the no explanation condition.

On each trial simulated online review information will be presented about two products that are described as “similarly-priced smartphone cases”. Products will be labeled by randomly generated letters. The mean online quality rating for each product will be shown by the number of stars and by a numerical value out of a maximum 5 (see Figure 1 for a screenshot of a target trial). The number of reviews on which ratings are based appears beside the star rating. Participants will be asked to rate their preference for one of the products on a 6-point scale. The left-right position of the more-reviewed product will be counterbalanced across trials. A total of 33 test trials will be presented in random order.

As noted in the Design Plan, performance on the 4 attention check trial items will be used to determine inclusion in the final analysis sample. Any individual who fails two or more attention check times will be excluded.